# MRTMOS – Week 3

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**Disclaimer:**
This has been written completely using my own knowledge, reputed literature and websites, and overwhelmingly, class notes of courses we took.
AI has only been used to fact-check and write-check at the end.

# Agenda

- Which topologies to implement?
  - Which ones to skip?
  - Why?
    - Math + reasoning
- Which elements should be off-chip or on-chip?
  - Why?
    - Math + reasoning

# Implementation of Topologies

# Assumptions and convention

- For single-input amplifiers, the transconductance $G_m = I_{out}/V_{in}$ would apply.
- For 2-input amplifiers, the transconductance $G_m = I_{out}/(V_+ - V_-)$ would apply.
  - **$V_+$ and $V_-$ refer to the input signal, not the amplifier input terminals.**
  - $G_m$ would adjust and change sign, but a positive current means current charging the capacitor at the output (coming out of amplifier).
  - This is for algebraic simplicity.
  - This is under the condition that the amplifier has only one output.
- At every node, KVL and KCL will prevail along with the sign convention.

# Schematics for all the stages

- **All Stages and Feedforward:** 5-T OTA
  - We can adjust parameters (width, length, tail current) depending on the use-case!
  - It can be used to invert as well as non-invert!
  - Only 5 transistors needed!
- **Voltage buffer:** Source follower
  - Noninverting, reliable buffer

# Why we picked them?

- **5-T OTA  for all stages and feedforward:**
  - Standardized topology makes layout reuse and verification effortless.
  - Dual inputs allow toggling between inverting and non-inverting.
  - Self-biasing active load and tail current source eliminates complex feedback circuits (common mode feedback circuitry in this case).
  - Decent (roughly 30-40dB) gain via high output resistance with low DC headroom.
  - Uniform architecture simplifies much of our work and student's work!
  - Easy to scale output driving current, gain etc. by adjusting transistor parameters.
    - by scaling W proportionally with $I_D$'s while holding $V_{ov}$ fixed via $g_m/I_D$.
- **Common drain amplifier as voltage buffer:**

$$Z_{in} \approx \frac{1}{j\omega[C_{gd} + C_{gs}(1 - A_v)]}$$

  - Insulated gate provides very high input impedance, protecting stage gain.
    - at AC, $C_{gs} \approx 30fF$ gives $|Z| \approx 5.3M\Omega$ at 1MHz, still $>> R_{out}$ which is of the order of $50\,k\Omega$
  - Blocks the feedforward current path through the Miller cap connected in series, kills the RHP zero and preserves the feedback path (as output faces the high input impedance and the intermediate output faces the low output impedance of the buffer).

# Some alternatives that came to our mind...

- **Common Source Amplifier for stage 3:**
  - **Temptation:**
    - **Maximum slew rate:** Lacks a tail current bottleneck, pulling massive transient surges directly into the capacitive load.
    - **Ultimate voltage swing:** Only two stacked transistors between supply rails allows near rail-to-rail output.
  - **Why 5-T OTA wins over it?**
    - **Polarity lock-in:** Common Source is permanently inverting, meaning that we won't be able to realize an amplifier in noninverting configuration, when we need it for nested and reverse-nested miller compensation.
- **Folded Cascode OTA (Cascode based circuits in general):**
  - **Temptation:**
    - **Massive single-stage gain:** Stacking transistors multiplies output resistance $(Z_{out} \approx g_m r_o^2)$, for the middle stage (the engine of the 3-stage amplifier). This further boosts the gain.
    - **Superior input range:** Allows input signals to swing close to the rails without pushing devices out of saturation.
  - **Why 5-T OTA wins over it?**
    - **Headroom starvation:** Folded cascode improves headroom over telescopic but still costs more than plain 5T-OTA, plus needs an added folding-bias branch (extra power/complexity).
    - **Biasing fragility:** Requires 3-4 precisely matched bias voltages, where any mismatch carries a potential risk to push cascode devices out of saturation given the already-tight margins

# Some alternatives that came to our mind...

- **Resistive load common source differential pair with tail current**
  - **Temptation:**
    - **Perfect linearity:** Resistors provide frequency-independent, highly linear load impedance.
  - **Why 5-T OTA wins over it?**
    - **Ohm's law trap:** Pushing Stage 3's 1 mA tail current through a 20 $k\Omega$ resistor demands 20 V of headroom. Impossible on our chip. If we reduce the current, it would place a bottleneck on the slew rate. Shrinking the resistor to fit a 3.3 V supply drops output impedance so low that voltage gain degrades sharply as a consequence of output impedance degradation.
- **Current load common source differential pair with resistance in tail.**
  - **Temptation:**
    - **Built-in linearity:** Tail resistor adds local negative feedback (source degeneration), stabilizing transconductance against process shifts.
    - **No bias distribution:** Eliminates routing a precise DC bias voltage $(V_{bias})$ to the bottom of the stage.

# Some alternatives that came to our mind...

- **Current load common source differential pair with resistance in tail.**

$$A_{dm} = \frac{1}{2} g_m R_D \qquad A_{cm} = \frac{-g_m R_D}{1 + 2 g_m R_{tail}} \qquad A_{cm} \approx -\frac{R_D}{2R_{tail}}$$

  - **Why 5-T OTA wins over it?**
    - **CMRR destruction:** Good noise rejection requires infinite tail impedance; a physical resistor lets supply noise bleed into the signal.
    - **Throttled slew rate:** A very large tail resistor sized to keep quiescent current low cannot supply more than $(V_{supply}-2V_{ov})/R_{tail}$ (or even more if we use a resistive load) during large-signal slewing without exceeding available headroom.
      - This caps achievable $I_{tail}$ below the $SR=I_{tail}/C_c$ target, unlike an active current source (~70kΩ output impedance) which holds current roughly constant regardless of voltage, up to compliance.
- **Class AB push-pull output:**
  - **Temptation:**
    - **Small quiescent current:** Burns very less DC current while resting, but opens floodgates during transient load demands.
    - **Thermal efficiency:** Eliminates burning a continuous, chip-heating 1 mA+ DC tail current just to maintain slew rate. Slew rate is decoupled to the tail current here because the path from load to Vdd/ground is the only path conducting current.
  - **Why 5-T OTA wins over it?**
    - **Thermal runaway:** Heating increases temperature, which decreases $V_{th}$, which increases $V_{ov}$, which increases quisscient current, which increases power dissipation, which increase temperature further.
      - **At high quisscient current and overdrives, when both devices are ON simultaneously, they conduct from supply to ground.**
    - **Single I/O:** We can’t connect it in both inverting and noninverting configuration. We need both to realize all our experiments.

# Some alternatives that came to our mind...

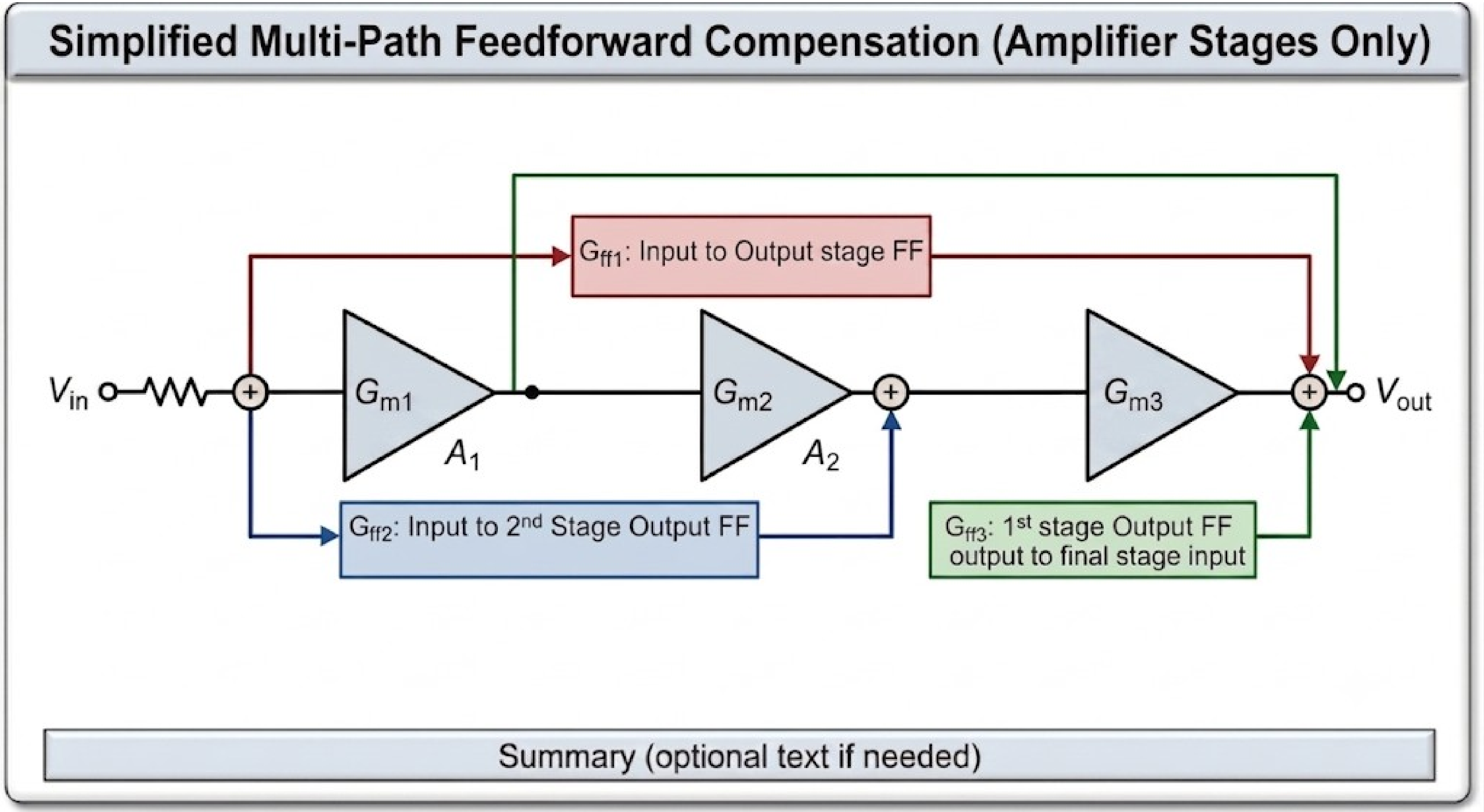
- **5-T OTA with unity-gain feedback (for the Miller voltage buffer):**
  - **Temptation:**
    - **Topology reuse:** We already have the 5-T OTA block designed, biased, and verified. Copy-pasting it as our buffer saves layout and verification time.
    - **Precision voltage tracking:** The high open-loop gain of the 5-T OTA guarantees the closed-loop buffer tracks the voltage almost perfectly $(A_v \approx 1, A_{cl} = 1\text{-}1/A_{ol})$, better than Source Follower.
  - **Why Source Follower wins over it?**
    - **The "Loop-in-a-Loop" danger:** The Miller feedback path must be very fast to stabilize the main amplifier. A 5-T OTA introduces its own internal poles, adding severe phase lag into the compensation path and destroying the main phase margin.
    - **Self-stability nightmare:** A 5-T OTA configured in unity-gain requires its own compensation to prevent oscillation. A source follower is inherently broadband and significantly stable.
    - **Area and power bloat:** Using 5 transistors and a dedicated tail current source just to block feedforward current is an overkill. A 1-2 transistor Source Follower provides the exact unidirectional isolation we need with a fraction of the silicon area and capacitance.

# The ones to keep: Standard topologies (No debate!)

- Dominant pole – shunt cap compensation
- Standard Miller compensation
- Miller + nulling resistor
- Nested Miller
- Reverse Nested Miller

# The ones that required some analysis

# Circuit Diagram



# Input to Output feedforward

$$\frac{V_{out}}{V_{in}} = [\frac{G_{ff} \cdot R_{out,ff}}{1 + \frac{s}{C_{ff} \cdot R_{out,ff}}} + \left(\frac{G_{m1} \cdot R_{out1}}{1 + \frac{s}{C_1 \cdot R_{out1}}}\right) \cdot \left(\frac{G_{m2} \cdot R_{out2}}{1 + \frac{s}{C_2 \cdot R_{out2}}}\right) \cdot \left(\frac{G_{m3} \cdot R_{out3}}{1 + \frac{s}{C_3 \cdot R_{out3}}}\right)]$$

- $G_{m3}$ factor would come only if it is 3 stage. Else omit it.
- When you simplify, you'll get $s^2$ term in numerator. 2 zeroes in 3-stage opamp! 1 zero in 2-stage!
- Calculating the dominant zero $(s_{z1})$ and non-dominant zeroes ($s_{z2}$,$s_{z3}$) (assumption: Main path gain >> feedforward path gain, 3-stage):

$$s_{z1} \approx -\frac{1}{C_{ff} \cdot R_{out,ff}}$$

$$s_{z2,3} = \pm j\sqrt{\frac{G_{m1}G_{m2}G_{m3} \cdot C_{ff}}{G_{ff} \cdot C_1 C_2 C_3}}$$

(3-stage)

$$s_{z,2} = \frac{G_{m1}G_{m2} \cdot C_{ff}}{G_{ff} \cdot C_1 C_2}$$

(2-stage)

- **Include it!**

# Input node to output of 2nd stage amplifier (Feedforward: 3-stage OTA)

- 3rd stage would add a pole as normal

$$V_2 = \left\{ V_{in} \cdot \left( \left[ \frac{G_{m1} \cdot R_{out1}}{1 + s \cdot C_1 \cdot R_{out1}} \right] \cdot G_{m2} \right) + V_{in} \cdot G_{ff} \right\} \cdot \left[ \frac{R_{out2}}{1 + s \cdot C_2 \cdot R_{out2}} \right]$$

- Calculating the Zero:

$$s = -\frac{G_{m1} \cdot G_{m2} \cdot R_{out1} + G_{ff}}{G_{ff} \cdot C_1 \cdot R_{out1}}$$

  - The numerator has a LHP zero provided that $G_{m1}, G_{m2}$ and $G_{ff}$ are in noninverting (positive) configuration.
  - $G_{ff}$ should NOT be negative, else the zero would be propelled to the RHP.
  - If it is so, either one of $G_{m1}, G_{m2}$ should be positive but the product of both should be negative, to make the numerator negative as well.

# Input of 2nd stage amplifier to output (Feedforward: 3-stage OTA)

- Output node:

$$V_{OUT} = \left\{ V_1 \cdot \left( \left[ \frac{G_{m2} \cdot R_{out2}}{1 + s \cdot C_2 \cdot R_{out2}} \right] \cdot G_{m3} \right) + V_1 \cdot G_{ff} \right\} \cdot \left[ \frac{R_{out3}}{1 + s \cdot C_3 \cdot R_{out3}} \right]$$

- Calculating the Zero:

$$s = -\frac{G_{m2} \cdot G_{m3} \cdot R_{out2} + G_{ff}}{G_{ff} \cdot C_2 \cdot R_{out2}}$$

  - The numerator has a LHP zero provided that $G_{m2}, G_{m3}$ and $G_{ff}$ are in noninverting (positive) configuration.
  - $G_{ff}$ should NOT be negative, else the zero would be propelled to the RHP.
  - If it is so, either one of $G_{m2}, G_{m3}$ should be positive but the product of both should be negative, to make the numerator negative as well.

# Gm followed by Miller in series (2-stage amplifier)

$V_1: \quad G_{m1}V_{in} + V_1\left(\frac{1}{R_{out1}} + sC_1\right) = 0 \qquad V_x: \quad G_{mff}V_1 = V_x\left(\frac{1}{R_{outff}} + sC_{ff}\right) + (V_x - V_{out})sC_m \qquad V_{out}: \quad (V_x - V_{out})sC_m = G_{m2}V_1 + V_{out}\left(\frac{1}{R_{out2}} + sC_2\right)$

- Solving the KCL equations above, the transfer function and the dominant pole, RHP zero:

$$H(s) = \frac{V_{out}}{V_{in}} = \frac{G_{m1}G_{m2}R_{out1}R_{out2} \cdot \left(1 - sR_{outff}\left[\frac{G_{mff}}{G_{m2}}C_m - C_{ff} - C_m\right]\right)}{(1 + sR_{out1}C_1)\cdot(1 + s(R_{outff}(C_{ff} + C_m) + R_{out2}(C_2 + C_m)) + s^2(R_{out2}R_{outff}(C_2C_{ff} + C_2C_m + C_{ff}C_m)))}$$

$$z_1 = \frac{1}{R_{outff}\left(C_{ff} + C_m\left[1 - \frac{G_{mff}}{G_{m2}}\right]\right)} \qquad p_1 = -\frac{1}{R_{out1}C_1}$$

- See the quadratic in the denominator, we can write it as $bs^2 + as + 1$. Solving it using discriminant:

$$p_{2,3} = \frac{-\left[R_{outff}(C_{ff} + C_m) + R_{out2}(C_2 + C_m)\right] \pm \sqrt{\left[R_{outff}(C_{ff} + C_m) + R_{out2}(C_2 + C_m)\right]^2 - 4\left[R_{out2}R_{outff}(C_2C_{ff} + C_2C_m + C_{ff}C_m)\right]}}{2R_{out2}R_{outff}(C_2C_{ff} + C_2C_m + C_{ff}C_m)}$$

- The value of the dominant pole is independent of $C_m$ unlike in normal Miller compensation!
- The value of RHP zero can be altered independent of dominant pole, by altering $G_{mff}$! We can push it into LHP!
  - **Complex Pole Risk:** If the discriminant is negative, both poles form a complex conjugate pair.
  - **Control Theory Map:** Comparing $bs^2 + as + 1 = 0$ to the standard form $s^2/\omega_n^2 + 2\zeta s/\omega_n + 1 = 0$ yields a damping ratio of $\zeta = a/2\sqrt{b}$.
  - **Physical Consequence:** If $\zeta < 1$, the amplifier is underdamped, resulting in severe transient ringing and peaking in the time domain.

## Capacitor following Gm - Nested Miller (3-stage amplifier)

- Similar thing as 2-stage Miller
- The crux is that Miller introduces an RHP zero and shifts the nondominant pole, if it has a Gm before it. The dominant pole is invariant.
- The magnitude would scale, but the underlying logic is the same.
- There is a potential to shift the zero to LHP!

# Should R/C be on/off chip?

# Resistor

- Nulling resistors must be placed off-chip.
- Students sweep to empirically discover the exact value.
- On-chip resistors lack any tuning capabilities.
- Wiper resistance uncertainty stays below 5% worst-case.
- Fixed precision metal-film resistors serve secondary verification.
- Precision resistors provide an ultra-tight 0.1% tolerance.
- On-chip resistor variations degrade matching by 20%.
- Off-chip components close the hand-analysis loop.
- Another option: Potentiometers enable continuous sweeping of $R_{\mathrm{null}}$.

# Capacitor

- All compensation capacitors must be off-chip components.
- On-chip MOM density is roughly $2\mathrm{fF}/\mu\mathrm{m}^2$ .
- Target $C_c = G_{m1}/(2\pi \cdot \mathrm{GBW}) \approx 318\ \mathrm{pF}$ .
- Area needed is $159{,}000\mu\mathrm{m}^2$ per capacitor.
- Total available chip area is $600{,}000\mu\mathrm{m}^2$.
- On-chip integration is physically completely impossible given that there are other components as well.
- Off-chip C0G dielectric guarantees $\pm 0.1\%$ tolerance.
- On-chip process variations suffer a $\pm 20\%$ error.
- Precise values allow exact experimental verification.

# The final verdict

- Keep it off-chip!
- Let students decide the values and plug them in!

# A Problem: Breadboard parasitics

- Breadboards are notoriously hostile to high-frequency analog integrated circuits.
- The physical structure of a standard breadboard introduces roughly 2 pF to 5 pF of hidden parasitic capacitance between adjacent rows.
- Furthermore, standard jumper wires act as tiny antennas, introducing stray inductance and coupling ambient noise directly into sensitive paths. It is a major error for high frequency circuits.
- How does this jeopardize our compensation experiments?
- These invisible environmental parasitics artificially shift the dominant and non-dominant poles before a student even plugs in a compensation capacitor.
- An intentionally uncompensated, unstable OTA might not oscillate simply because the breadboard itself is accidentally acting as a compensation network!
- High-impedance internal nodes routed off-chip will suffer severe phase margin degradation entirely due to the test setup. at compensation node, even 5pF of stray capacitance creates a parasitic pole at 318kHz, within an octave of our 482kHz GBW target.
- The exact, predictable mathematical derivations we want students to empirically verify (like precisely shifting the RHP zero) become completely muddled by environmental noise.
- **The bottom line: Students must measure the exact physics of our silicon, not the physical flaws of a plastic breadboard.**

# Solution: PCB implementation (microcontroller-style board)

- We will bypass the breadboard entirely and design a fully dedicated, professional-grade evaluation PCB.
- This shifts the project from a fragile "student chip" into a robust, self-contained educational microcontroller-style platform.
- To tackle the parasitic capacitance, we must execute a careful ground plane strategy.
- The Trap: Completely removing the ground plane beneath the entire SoC is dangerous; it destroys the return paths for digital SPI control logic, causing digital switching noise to radiate directly into our analog stages.
- The Real Solution: We will maintain a solid ground plane for power delivery and digital logic, but utilize highly localized "copper keep-outs" (voids).
- We will strictly remove all ground and power planes only directly beneath the sensitive high-impedance routing traces and the female plug-in receptacles.
- **We will use our knowledge and experience in PCB design and routing :)**

# Hardware and pedagogical features

- For the off-chip component interface, standard Arduino-style square headers are too loose for thin passive leads.
- We will use Machine-Pin (SIP) sockets: Their round, gold-plated inner contacts tightly grip standard C0G capacitors and metal-film resistors, guaranteeing a low contact resistance, around 10-50mΩ, negligible against circuit impedances.
- The Silkscreen as a Syllabus: We will draw the simplified op-amp block diagrams and routing paths directly on the top-layer silkscreen of the board.
- By explicitly labeling the component sockets (e.g., "$C_c$ Node", "$R_{null}$ Node") right on the board, the PCB itself acts as the student's lab manual.
- Finally, we can use BNC or SMA cables, and probes for signal I/O, ensuring perfectly shielded, low-noise interfacing with university oscilloscopes and function generators.

# Lab experiments

# Sequence and Methodology

- **Phase 1: Single-Stage Characterization (Small & Large Signal)**
  - Measure discrete amplifier blocks
  - Small signal to measure Gm, Large signal to measure swing and ICMR
  - Tail current to measure power $(P_{diss} = V_{DD} * I_{tail})$
- **Phase 2: Frequency response, compensation (100mV p-p input)**
  - Compare frequency response in uncompensated and various forms of compensation
  - Vary values of R/C or feedforward Gm in compensation networks, and plot graphs to develop intuition.
- **Phase 3: Characterizing multi-stage amplifiers (small & large signal)**
  - Measure the entire amplifier combined, steps similar to phase 1
- **Phase 4: Basic applications and system level applications**
  - Filter design

# Measuring and Characterization of single OTA: Gain and output impedance

- Students would connect the amplifier in negative feedback network configuration. Input voltage should be small (50mV peak-peak or so) at a reasonable frequency (5-10kHz).

- The OTA would be connected to a resistive load, and the voltage induced in the resistor would be fed back into the inverting terminal of the OTA. Input voltage would be connected to the noninverting terminal.

- By varying the resistive load $R_L$, students would in turn, vary the negative feedback gain.

$$A_v = \frac{V_{out}}{V_{in}} = \frac{G_m \cdot R_L||R_{out}}{1 + G_m \cdot R_L||R_{out}} \implies \frac{1}{A_v} - 1 = \left(\frac{1}{G_m}\right)\frac{1}{R_L} + \frac{1}{G_m R_{out}}$$

- Plotting $(1/A_v)$ -1 vs $1/R_L$ yields a straight line where the slope directly equals $1/G_m$. The y-intercept comes out to be $1/G_m \cdot R_{out}$.

- It is recommended to use small values of resistances (upto a few $k\Omega$) to first measure $G_m$ as a small value of $R_L$ would also push the output pole far away.

- At high values of $R_L$, we can end up measuring $R_{out}$!

- But at very high values of $R_L$, the output pole would occur at lower frequencies.

- After experiment is done, students would get the passive and transistor parameters and topologies of amplifiers, and measure via hand calculation!

# Measuring Gain and Characterization of the Voltage buffer

- It can be done in a straightforward way, as the voltage gain of the voltage buffer is constant and there is no fear of excess voltage and voltage gain deterioration depending on the load unlike in an OTA.
- Connect a sinusoidal voltage source at the input of the voltage buffer. 50mV peak-peak
- We can measure the output voltage and calculate transfer function.
- We can vary the frequency and amplitude to get an idea of the output swing and bandwidth.
- We can also get to know the ICMR, as we increase the input voltage and we can get to know deviation in output waveform (eg. Clipping) to identify ICMR limits.
- After experiment is done, students would get the passive and transistor parameters and topologies of amplifiers, and measure via hand calculation!

# Compensation Networks – Analysis and comparison

1) **Establish the Baseline:** Students utilize their previously extracted $G_m$ and $R_{out}$ values for the individual blocks.
2) **The Uncompensated Benchmark:** Students cascade the blocks into a 2- or 3-stage amplifier and measure the raw frequency response (identifying poles, zeros, DC gain, and phase/gain margins).
3) **Calculate, Then Connect:** Students mathematically design various compensation networks by hand before physically wiring them on the bench. Make the measurements as in step 2.
4) **Observe and Compare:** Students populate a master observation table to directly compare how each compensation topology shifts the parameters and stabilizes the system.

# Analysis and Characterization of multi-stage amplifiers

- Students would connect the multi-stage OTA in negative feedback network configuration. Input voltage should be small (50mV peak-peak or so) at a reasonable frequency (5-10kHz). **Don't connect any compensation network yet for baseline measurements.**

- The OTA would be connected to a resistive load, and the voltage induced in the resistor would be fed back into the inverting terminal of the OTA. Input voltage would be connected to the noninverting terminal.

- By varying the resistive load $R_L$, students would in turn, vary the negative feedback gain.

$$A_v = \frac{V_{out}}{V_{in}} = \frac{G_m \cdot R_L||R_{out}}{1 + G_m \cdot R_L||R_{out}} \implies \frac{1}{A_v} - 1 = \left(\frac{1}{G_m}\right)\frac{1}{R_L} + \frac{1}{G_m R_{out}}$$

- Plotting $(1/A_v)$-1 vs $1/R_L$ yields a straight line where the slope directly equals $1/G_m$. The y-intercept comes out to be $1/G_m \cdot R_{out}$.

- It is recommended to use small values of resistances (upto a few $k\Omega$) to first measure $G_m$ as a small value of $R_L$ would also push the output pole far away.

- Compare the experimentally extracted overall parameters $(G_m, R_{out})$ of the complete multi-stage amplifier against the individual stage parameters previously measured in phase one. Verify the theoretical mathematical relationships of the cascaded stages.

# Vary compensation network elements, develop intuition

- **Calculate by Hand:** Derive the transfer function $A_v$ to map out target pole and zero locations.
- **Connect the Topology:** Link the multi-stage amplifier network to your chosen compensation loop configuration.
- **Inject the Step Response:** Vary R,C and feedforward $G_m$ values, and apply a step response to observe the resulting waveforms, and observe the I/O plots and make an observation table.
- **The Transient-to-Control Theory Map:**
  - **Sluggish Rise (No Overshoot):** Overdamped response ($\zeta > 1$) with excessive Phase Margin ($\varphi_m > 76^\circ$) but choked bandwidth.
  - **Ringing & Peak Overshoot:** Underdamped condition ($0 < \zeta < 1$); non-dominant poles have crept inside the loop bandwidth, degrading Phase Margin ($\varphi_m < 60^\circ$ typically).
  - **Sustained Oscillations:** Total system instability where Phase Margin hits $0^\circ$ and loop poles have migrated into the unstable Right-Half Plane (RHP).
  - **The "Slow Creeping Tail":** Diagnostic signature of a Pole-Zero Doublet; imperfect pole-zero cancellation traps an agonizingly slow exponential time constant in your settling path.
  - **Initial Reverse-Direction "Dip":** Classic Non-Minimum Phase behavior caused by a dominant Right-Half Plane (RHP) Zero momentarily outrunning the main amplifier path.
- **First do the calculations and predict the waveforms, then do it practically. Don’t be a simulator monkey!**

# System level applications 1: Design a filter!

- You have the specifications of all the amplifiers (discrete and combined!)
- Most amplifiers have isolated terminals.
- Design a filter with 1-2 amplifiers:
  - State variable, Multiple feedback, Sallen Key, Biquad, simple amplifier based active filters.
  - Bandwidth is an important aspect in filter design, and poles/zeroes and therefore, compensation play a significant role.
- First calculate the transfer function by hand.
- Then form the circuit and connect.
- After that, you can verify the measurements with experimental results.

# System level applications 2: Active inductor!

- You have the specifications of all the amplifiers (discrete and combined!)
- Most amplifiers have isolated terminals.
- Design an active inductor with 1-2 amplifiers:
  - Various literature is available
  - Active inductor input impedance is heavily influenced by $G_m$ of the amplifiers, the capacitance of its output node, which is also a pole.
  - Pole compensation indirectly means impedance alteration!
- First calculate the transfer function by hand.
- Then form the circuit and connect.
- After that, you can verify the measurements with experimental results.

# System-Level Design: The Compensation Challenge (Application based Course project)

- **The Objective**
  - Real-world applications break standard amplifiers. You will be assigned strict system-level specifications. Your task is to diagnose exactly why a standard topology fails under these conditions and implement an advanced compensation strategy to save it.
- **The Engineering Workflow**
  - **Step 1: Diagnose the Failure:** Test a baseline amplifier against your assigned specs to find the exact instability mechanism (e.g., ringing, oscillation, or slow settling).
  - **Step 2: Strategize the Fix:** Select an advanced compensation architecture tailored directly to the failure mechanism.
  - **Step 3: Validate the System:** Prove your newly compensated circuit survives across extreme load conditions and process corners.
- **Target Application Scenarios**
  - **Power & Display Drivers:** Stabilizing the amplifier against massive, unpredictable capacitive loads.
  - **Precision ADC Cores:** Eliminating "slow creeping" settling errors to meet strict high-speed clock deadlines.
  - **RF/Wideband Systems:** Bypassing internal bottlenecks to achieve massive bandwidth without destroying phase margin.

# Thank You :)

Questions ?